Sahaj Bhadja

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Summary

Software Engineer with 2+ years building **Python** and **TypeScript** backend APIs, **data pipelines**, and **PostgreSQL** query tuning, plus production LLM applications with per-call usage logging, eager to grow in agent analytics.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Database Systems, Analysis of Algorithms, Advanced NLP, Web Technologies

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Data Retrieval, Operating Systems, Software Engineering, Object-Oriented Programming

Experience

USC Dornsife **September 2025 – Present**
Software Engineer *Los Angeles, CA*

- Investigated slow dashboard queries on a multi-tenant **PostgreSQL** and Azure SQL platform, redesigning indexes and access patterns to cut latency from 1.2s to 350ms for 500+ users.
- Built **data pipelines** and background jobs with FastAPI, validation, and reconciliation checks, running 5K+ monthly integration jobs at 99%+ reliability and processing 1K+ LLM jobs at 3x throughput.

Grad8 **June 2025 – August 2025**
Software Engineer Intern *Los Angeles, CA*

- Built a production **LLM application** with React and Flask that chains rubric generation, OCR, grading, and feedback across GPT-4o, Claude, and Qwen-VL for 150+ instructors and 10K+ daily requests.
- Logged every model call with model name, feature, token counts, and estimated cost in normalized **MySQL** tables, enabling per-model usage and cost analysis while cutting feedback turnaround 60%.

Aasma Technology Solutions **August 2023 – July 2024**
Software Developer Engineer – SecureSync *India*

- Redesigned **ETL** pipelines on AWS Glue and S3 that normalized raw JSON, CSV, and Parquet activity logs from multiple sources for analytics and API consumption, boosting pipeline efficiency 68%.
- Wrote 20+ **unit and integration test** suites with PyTest and Mockito in nightly Jenkins CI/CD, and optimized REST APIs with pagination, field filtering, and caching, improving response times 30%.

Aasma Technology Solutions **May 2023 – August 2023**
Full Stack Developer Intern – VibrantChem *India*

- Built eight TypeScript and Node.js microservices for resources, billing, and leads with React and MongoDB, exposing **REST APIs** and real-time WebSocket updates across 2M+ records synchronized between locations.
- Analyzed Google **Analytics** product data on feature usage, load times, and drop-off points and debugged services with Kibana logs and Sentry, guiding changes that raised lead conversions 15%.

LST Capitals **November 2022 – January 2023**
Web Developer Intern *India*

- Built hourly Databricks and Spark **event pipelines** over 50K+ booking and payment events, aggregating peak hours, most-used services, daily active users, and revenue for internal dashboards and client reports.
- Developed Python Flask APIs with Firebase real-time sync and idempotent payment **webhooks**, sustaining sub-200ms latency and 40% faster booking confirmations while cutting manual refund handling by 70%.

Skills

Languages: Python, TypeScript, JavaScript, SQL, Java, C++
Backend: FastAPI, Flask, Node.js, Express.js, REST APIs, WebSockets, Background Jobs
Data: PostgreSQL, Query Optimization, Indexing, SQL Server, MongoDB, Databricks, Spark, ETL Pipelines, Data Validation
AI & Analytics: LLM Applications, Prompt Chaining, RAG, Embeddings, Google Analytics, Funnel Analysis
Cloud & Tooling: AWS, Azure, GCP, Docker, CI/CD, Git, Kibana, Sentry
Fundamentals & Testing: Data Structures, Algorithms, Unit Testing, Integration Testing, PyTest, Code Reviews

Projects

LLM Trace Evaluation Pipeline | *TypeScript, Node.js, Python, ClickHouse, PostgreSQL, Claude*

- Designed a TypeScript and Python pipeline that ingests agent traces, scores task success and tool-call errors with an **LLM judge**, and loads sessions into **ClickHouse** behind a session-explorer API.

Chat2SQL Code | *Java, TypeScript, PostgreSQL, Snowflake, Docker*

- Built an AI-assisted SQL workspace on Java and TypeScript for 10+ databases including **PostgreSQL** and Snowflake, generating schema-aware analytical queries at 82% accuracy and AI-generated reporting dashboards.

Big-Brain Live | *LLaMA-2, FAISS, PostgreSQL, GCP, REST APIs*

- Built a cloud **ingestion and indexing pipeline** over 50K+ documents with incremental updates and PostgreSQL metadata, powering LLaMA-2 RAG at 95% query accuracy and cutting retrieval latency 40%.